

SUPPLEMENTARY APPENDIX

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Armstrong AJ, Azad AA, Iguchi T, et al. Improved Survival with Enzalutamide in Patients with Metastatic Hormone-Sensitive Prostate Cancer

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Investigators are listed in alphabetical order by country and last name. This table only includes investigators who consented to be acknowledged by name.

Statistical Analysis

Baseline characteristics and efficacy outcomes were evaluated in the intent-to-treat population; patients who crossed over from placebo plus ADT to enzalutamide plus ADT were analyzed as randomized. Safety was assessed in patients who received at least one dose of study drug. To adjust for treatment duration, TEAEs per 100 patient-years were calculated as: $\text{number of events} \times 100 / \text{sum of the treatment-emergent period duration for all patients in the corresponding treatment group}$. Median OS was estimated by Kaplan–Meier methods, with two-sided 95% CIs calculated by the Brookmeyer and Crowley method. HRs relative to placebo plus ADT were determined using Cox regression model stratified for prior docetaxel use and disease volume. Similar analyses were performed for other time-to-event endpoints. A total of 342 death events were required to provide 80% power to detect a target HR of 0.73 by stratified log-rank test at a two-sided alpha error of 0.04, based on an O’Brien–Fleming boundary and designated by a parallel testing strategy employed to control for type I error across the key secondary efficacy endpoints. To assess the potential impact of treatment crossover, a prespecified sensitivity analysis using the rank-preserving structural time failure model,¹ stratified by disease volume and prior docetaxel use, was performed.

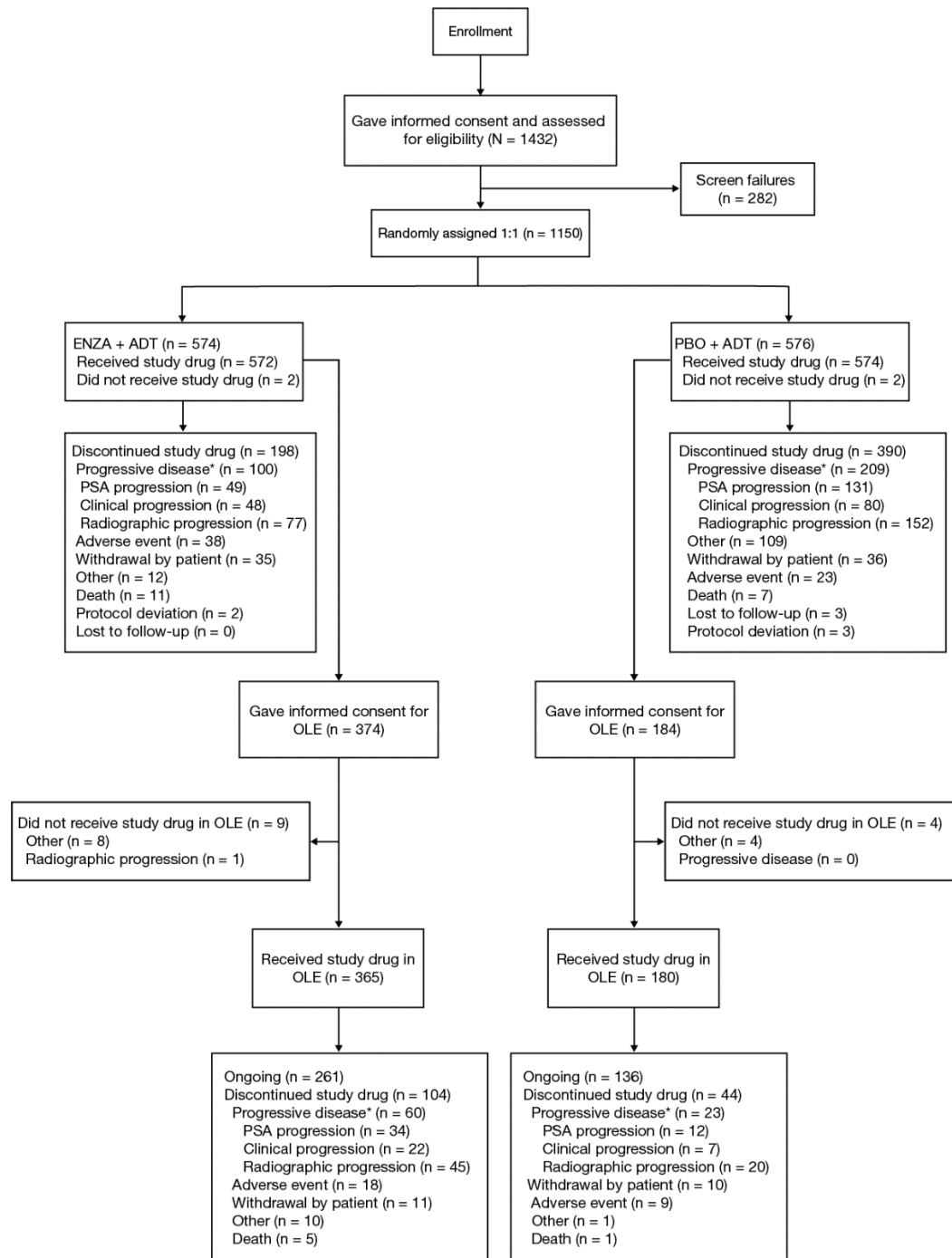
Subgroup analyses of OS were performed in prespecified patient populations based on age, geographic region, Eastern Cooperative Oncology Group performance status, Gleason score, disease location at baseline, baseline PSA value, prior local therapy (defined only as radical prostatectomy and/or radiation therapy of the prostate area), prior systemic treatment (docetaxel), and disease volume as defined by CHAARTED² (NCT00309985) criteria. The definition of soft tissue disease includes both lymphatic and visceral metastases for which clinical outcomes have been reported to differ in both mHSPC³ and castration-resistant disease.⁴ To better characterize

survival in patients with soft tissue disease, exploratory post hoc analyses on additional subgroups were performed. Subgroup analyses were not powered to compare treatment groups in individual subgroups.

All secondary endpoints underwent final formal statistical analysis at the time of the primary rPFS analysis. For descriptive purposes, we provide an updated analysis of rPFS (investigator assessed) and other secondary endpoints at the time of the data cutoff for the final OS analysis. For the analysis of time to subsequent antineoplastic therapy, the receipt of crossover enzalutamide by patients originally randomized to placebo plus ADT was not counted.

Limited data were imputed; missing start and stop dates for adverse events and medications were imputed as described in the statistical analysis plan for the open-label extension.

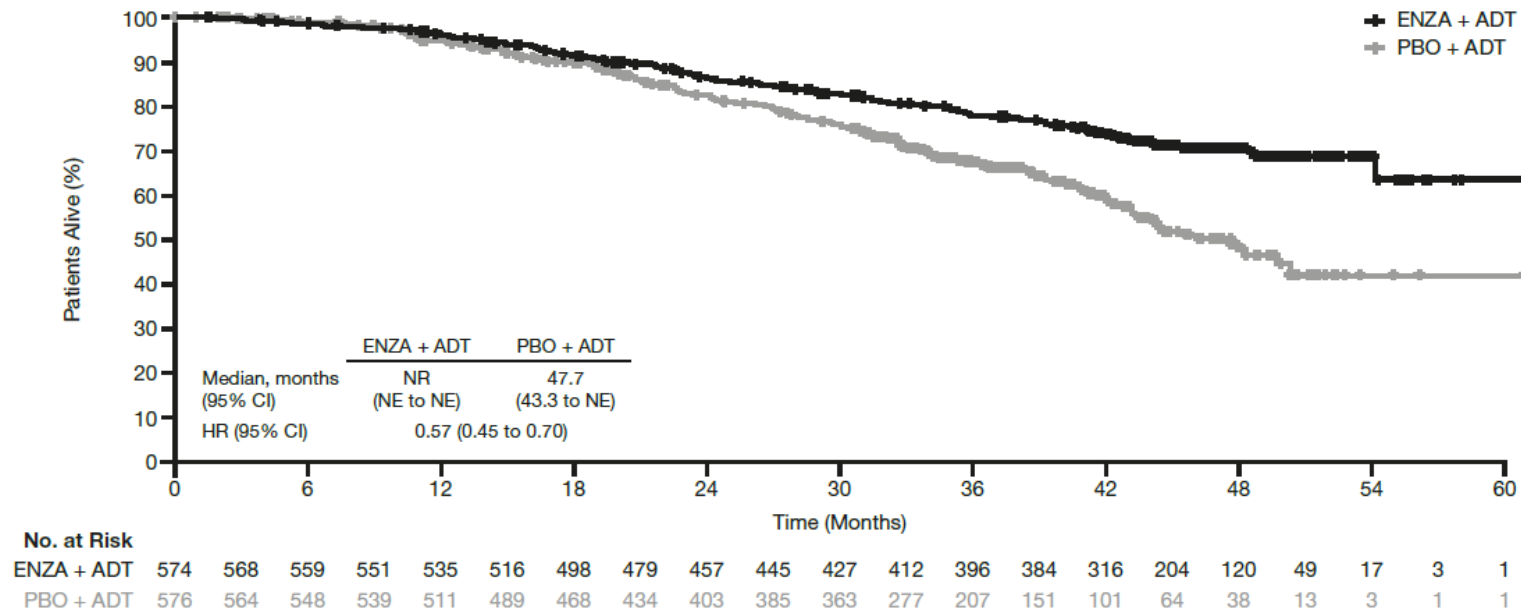
Figure S1. Patient Disposition in ARCHES



*A patient may be reported in multiple categories.

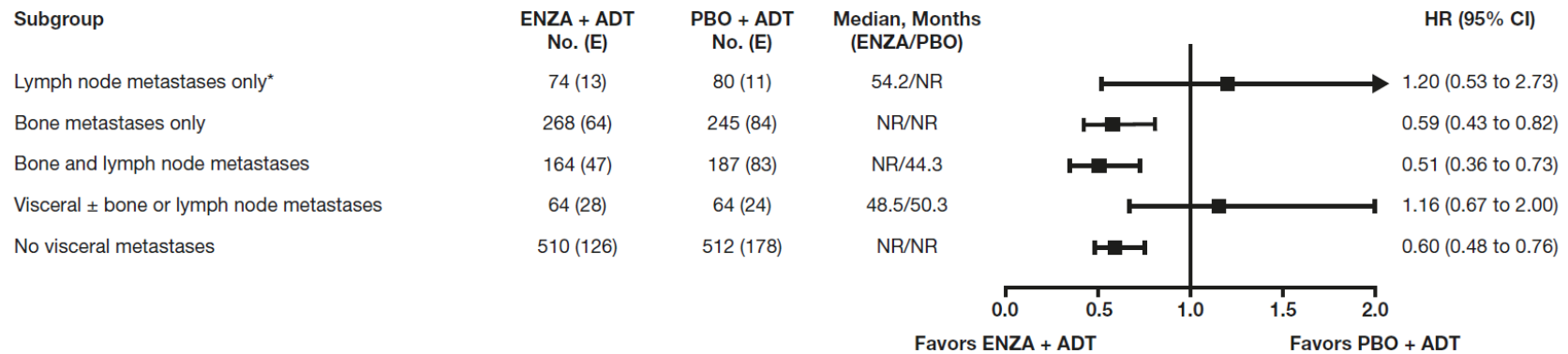
ADT, androgen deprivation therapy; ENZA, enzalutamide; OLE, open-label extension; PBO, placebo; PSA, prostate-specific antigen.

Figure S2. Kaplan–Meier Estimate of OS Adjusted for Crossover from Placebo plus ADT to Enzalutamide plus ADT Using RPSFT (Intent-to-Treat Population)



ADT, androgen deprivation therapy; CI, confidence interval; ENZA, enzalutamide; HR, hazard ratio; NE, not evaluable; NR, not reached; OS, overall survival; PBO, placebo; RPSFT, rank-preserving structural failure time.

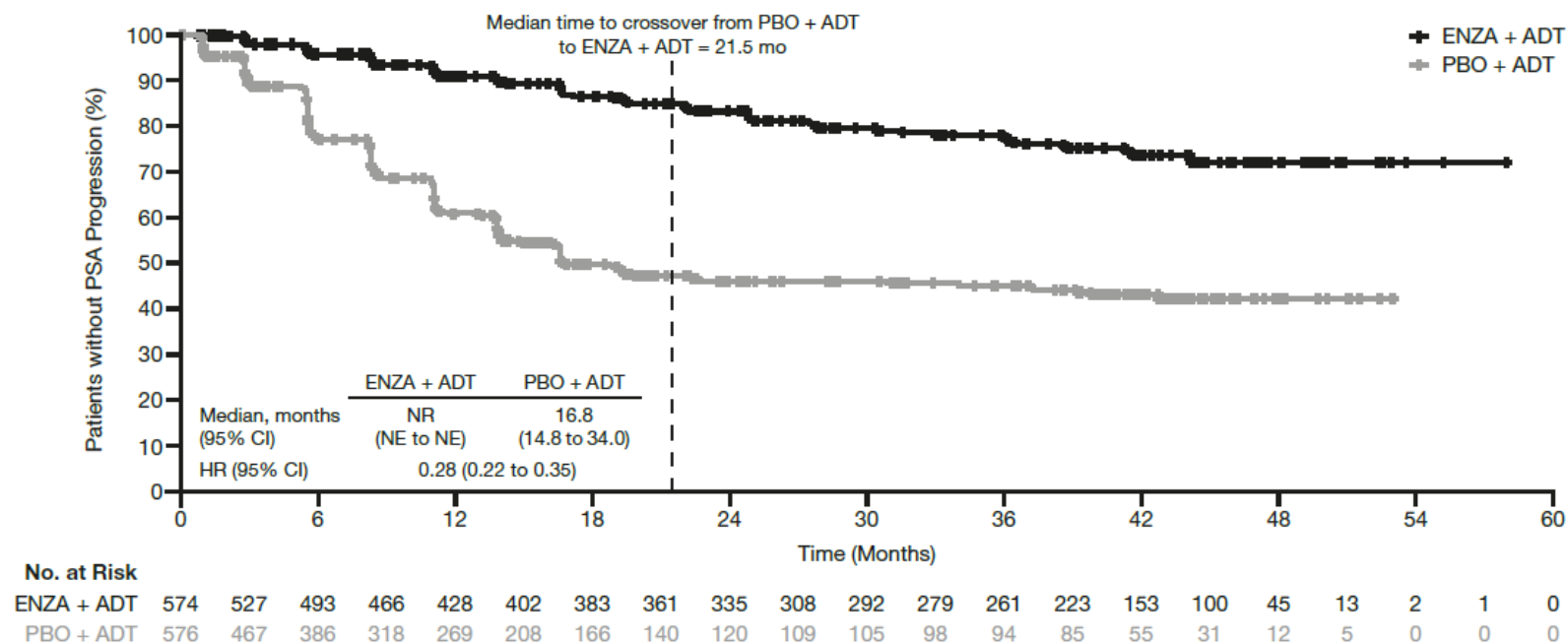
Figure S3. Forest Plot of OS Exploratory Post Hoc Subgroup Analyses (Intent-to-Treat Population)



*Patients with confirmed lymph node metastases or unconfirmed metastatic disease.

ADT, androgen deprivation therapy; CI, confidence interval; E, events; ENZA, enzalutamide; HR, hazard ratio; NE, not evaluable; NR, not reached; OS, overall survival; PBO, placebo.

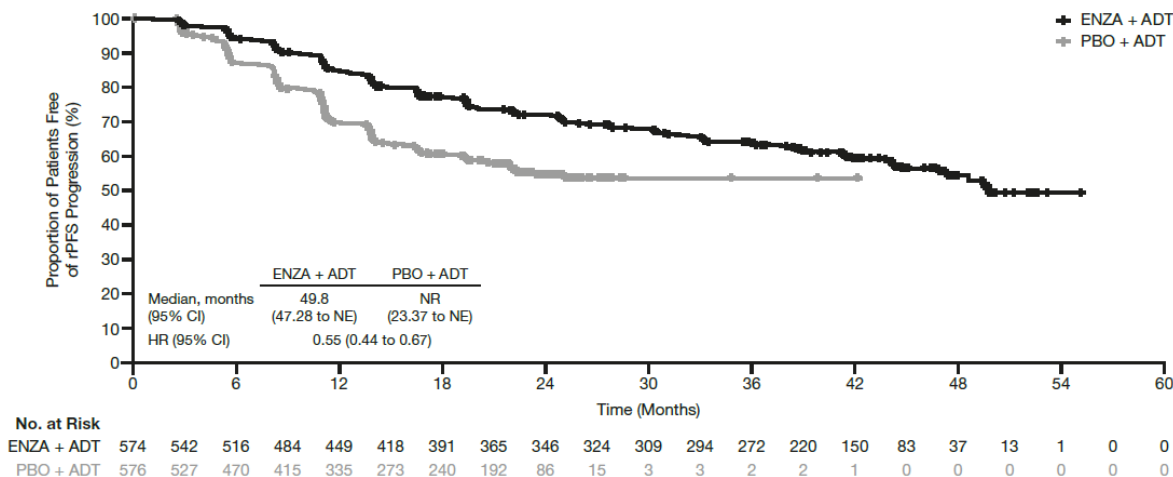
Figure S4. Kaplan–Meier Estimate of Time to PSA Progression (Intent-to-Treat Population)



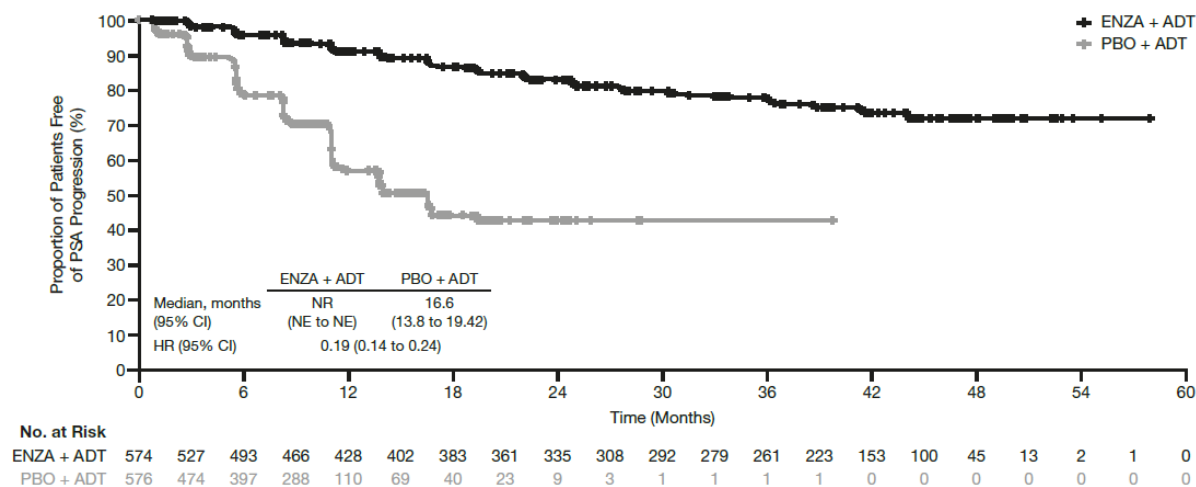
ADT, androgen deprivation therapy; CI, confidence interval; ENZA, enzalutamide; HR, hazard ratio; NE, not evaluable; PBO, placebo; PSA, prostate-specific antigen.

Figure S5. Kaplan–Meier Estimate of rPFS and Time to PSA Progression Adjusted for Crossover from Placebo plus ADT to Enzalutamide plus ADT Using RPSFT (Intent-to-Treat Population)

rPFS (Investigator Assessed) Adjusted for Crossover

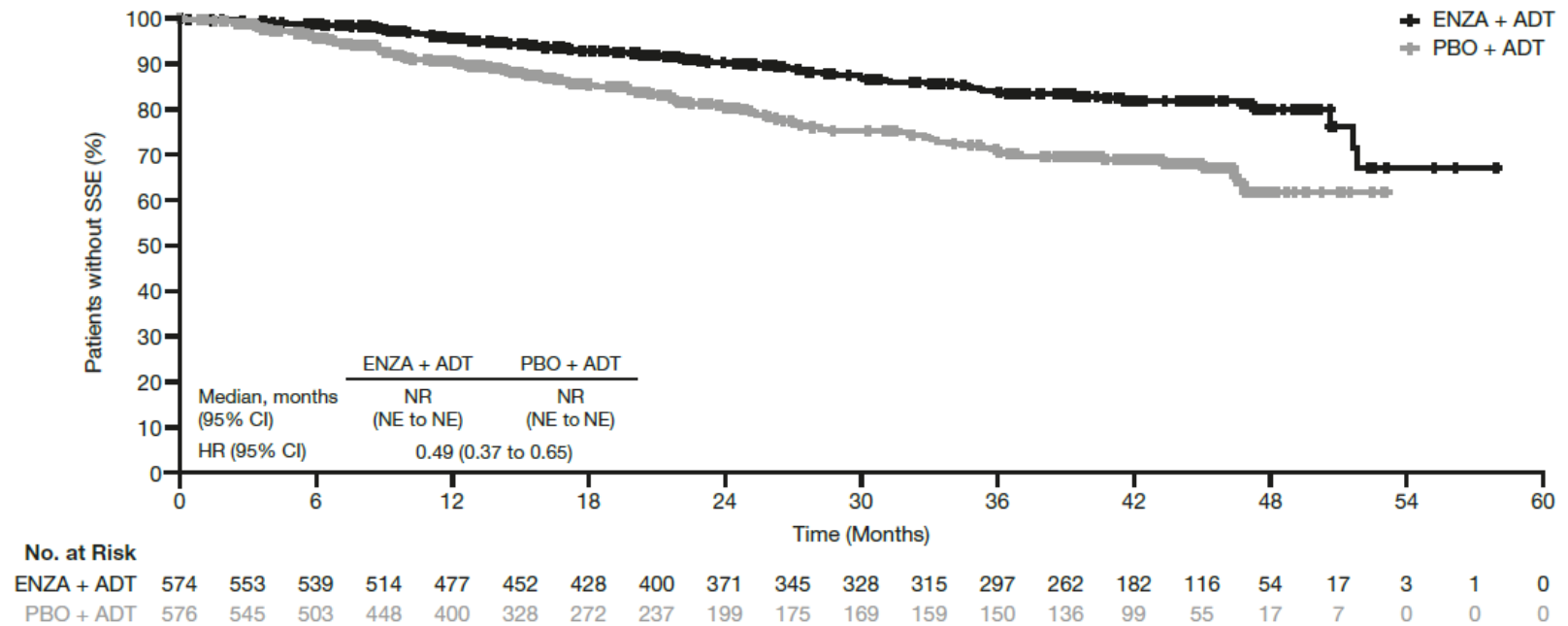


Time to PSA Progression Adjusted for Crossover



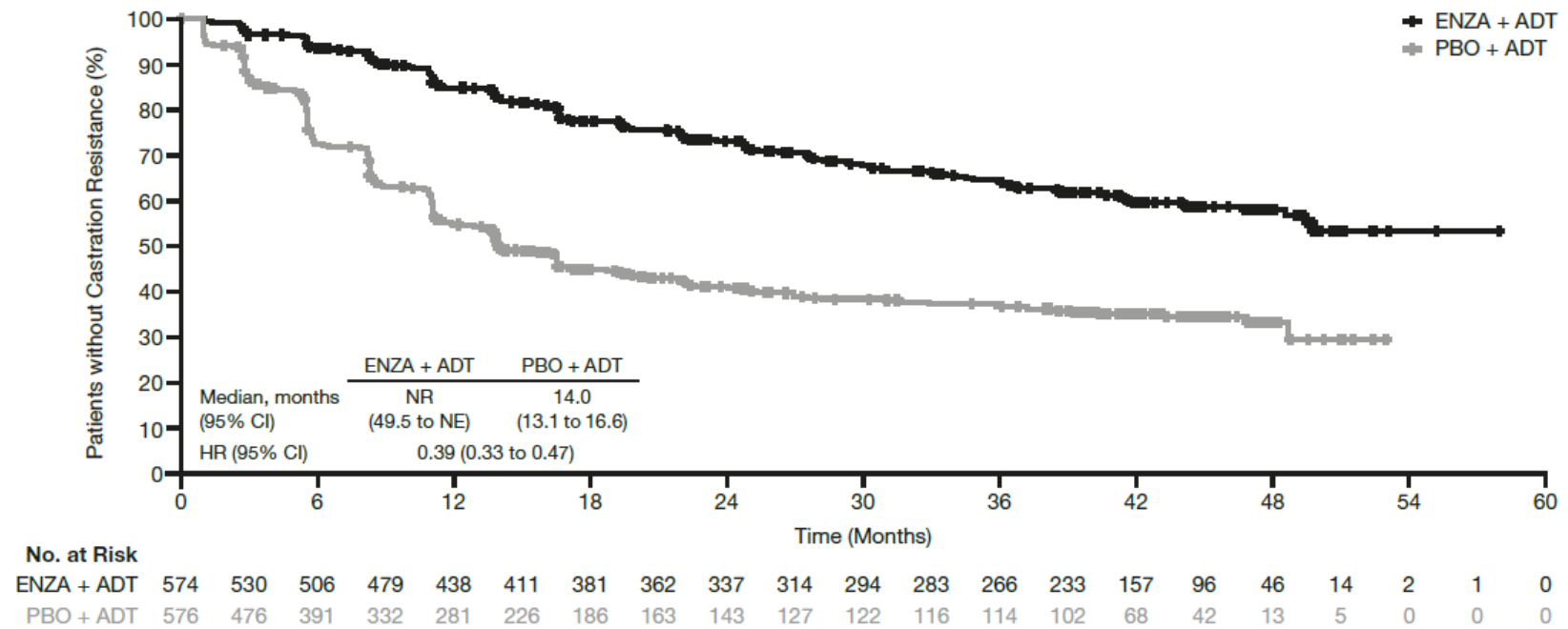
ADT, androgen deprivation therapy; CI, confidence interval; ENZA, enzalutamide; HR, hazard ratio; NE, not evaluable; NR, not reached; PBO, placebo; PSA, prostate-specific antigen; rPFS, radiographic progression-free survival; RPSFT, rank-preserving structural failure time.

Figure S6. Kaplan–Meier Estimates of Time to First SSE (Intent-to-Treat Population)



ADT, androgen deprivation therapy; CI, confidence interval; ENZA, enzalutamide; HR, hazard ratio; NE, not evaluable; NR, not reached; PBO, placebo; SSE, symptomatic skeletal event.

Figure S7. Kaplan–Meier Estimate of Time to Castration Resistance (Intent-to-Treat Population)



ADT, androgen deprivation therapy; CI, confidence interval; ENZA, enzalutamide; HR, hazard ratio; NE, not evaluable; NR, not reached; PBO, placebo.

Table S1. First Antineoplastic Therapy for Prostate Cancer after Treatment Discontinuation (Intent-to-Treat Population)

First Antineoplastic Therapy for Prostate Cancer, no. (%)*	ENZA + ADT (n = 574)	PBO + ADT (n = 576)	PBO crossover (n = 184)
Any	131 (22.8)	221 (38.4)	15 (8.2)
Docetaxel	48 (8.4)	71 (12.3)	5 (2.7)
Abiraterone acetate	26 (4.5)	42 (7.3)	3 (1.6)
Enzalutamide	7 (1.2)	61 [†] (10.6)	0
Bicalutamide/flutamide	8 (1.4)	23 (4.0)	1 (0.5)
Cabazitaxel	11 (1.9)	6 (1.0)	1 (0.5)
Sipuleucel-T	4 (0.7)	6 (1.0)	0
Radium	6 (1.0)	4 (0.7)	0
Other [‡]	23 (4.0)	12 (2.1)	6 (3.3)

*Patients may have started more than one subsequent antineoplastic therapy concurrently.

[†]Does not include enzalutamide received by patients who crossed over from PBO plus ADT to ENZA plus ADT (n = 180).

[‡]Includes blinded therapy, capecitabine, carboplatin, carboplatin with paclitaxel, cisplatin, combinations of antineoplastic agents,

darolutamide, degarelix, denosumab, dexamethasone, diethylstilbestrol, etoposide, folfox, investigational antineoplastic drugs, ipilimumab, irinotecan, mitoxantrone, nivolumab, olaparib, other alimentary tract and metabolism products, prednisone, and rucaparib.

ADT, androgen deprivation therapy; ENZA, enzalutamide; PBO, placebo.

Table S2. Efficacy Outcomes at the Time of the Final Analysis (Intent-to-Treat Population)

Endpoint	ENZA + ADT (n = 574)	PBO + ADT (n = 576)	HR (95% CI)	P
OS				
Median OS, months (95% CI)	NR (NE to NE)	NR (49.7 to NE)	0.66 (0.53 to 0.81)	<0.001
Median OS – RPSFT sensitivity analysis, months (95% CI)	NR (NE to NE)	47.7 (43.3 to NE)	0.57 (0.45 to 0.70)	<0.001
Median time to first antineoplastic therapy, months (95% CI)	NR (NE to NE)	40.5 (26.3 to NE)	0.38 (0.31 to 0.48)	
Median rPFS (investigator assessed), months (95% CI)*	49.8 (47.3 to NE)	38.9 (28.2 to 46.2)	0.63 (0.52 to 0.76)	
Median time to PSA progression, months (95% CI)	NR (NE to NE)	16.8 (14.8 to 34.0)	0.28 (0.22 to 0.35)	
Median time to deterioration of urinary symptoms, months (95% CI)†	33.4 (24.8 to 44.2)	19.6 (16.7 to 27.2)	0.88 (0.73 to 1.05)	

Median time to first SSE, months (95% CI)	NR (NE to NE)	NR (NE to NE)	0.49 (0.37 to 0.65)
Median time to castration resistance, months (95% CI)	NR (49.5 to NE)	14.0 (13.1 to 16.6)	0.39 (0.33 to 0.47)
Median time to deterioration of quality of life, months (95% CI) [‡]	11.6 (11.1 to 14.1)	11.1 (8.8 to 13.8)	0.92 (0.80 to 1.07)
Median time to pain progression, months (95% CI) [§]	8.5 (8.3 to 11.1)	8.3 (5.7 to 8.5)	0.96 (0.83 to 1.10)

*Radiographic progression was assessed by investigators according to Prostate Cancer Working Group 2 criteria.

[†]A deterioration in urinary symptoms was defined as an increase in the QLQ-PR25 modified urinary symptoms scale score by $\geq 50\%$ of the standard deviation observed in the QLQ-PR25 modified urinary symptoms scale score at baseline (i.e., Q31–Q33).

[‡]A deterioration of quality of life was defined as a decrease of ≥ 10 points in the total Functional Assessment of Cancer Therapy–Prostate score from baseline.

[§]Pain progression was defined as an increase of $\geq 30\%$ from baseline in the average Brief Pain Inventory–Short Form item scores.

ADT, androgen deprivation therapy; CI, confidence interval; ENZA, enzalutamide; NE, not evaluable; NR, not reached; PBO, placebo; PSA, prostate-specific antigen; rPFS, radiographic progression-free survival; RPSFT, rank-preserving structural failure time; SSE, symptomatic skeletal event.

Table S3. TEAEs (Safety Analysis Set)

TEAE, no. (%)*	ENZA + ADT		PBO + ADT [†]		PBO + ADT Crossover	
	(n = 572)		(n = 574)		(n = 180)	
Most common TEAEs (occurring in ≥10% of patients in any group)	All Grades	Grade 3–4	All Grades	Grade 3–4	All Grades	Grade 3–4
Hot flush	171 (29.9)	2 (0.3)	131 (22.8)	0	15 (8.3)	0
Fatigue	143 (25.0)	8 (1.4)	92 (16.0)	8 (1.4)	39 (21.7)	4 (2.2)
Arthralgia	104 (18.2)	2 (0.3)	65 (11.3)	5 (0.9)	12 (6.7)	1 (0.6)
Back pain	76 (13.3)	8 (1.4)	69 (12.0)	5 (0.9)	19 (10.6)	4 (2.2)
Hypertension	75 (13.1)	27 (4.7)	35 (6.1)	11 (1.9)	13 (7.2)	6 (3.3)
Falls	58 (10.1)	7 (1.2)	19 (3.3)	3 (0.5)	13 (7.2)	2 (1.1)
Most common study drug-related TEAEs (occurring in ≥5% of patients in any group)						
Fatigue	110 (19.2)	5 (0.9)	67 (11.7)	7 (1.2)	31 (17.2)	4 (2.2)

Hot flush	133 (23.3)	2 (0.3)	105 (18.3)	0	13 (7.2)	0
Hypertension	44 (7.7)	22 (3.8)	20 (3.5)	5 (0.9)	7 (3.9)	3 (1.7)
Increased weight	36 (6.3)	1 (0.2)	25 (4.4)	1 (0.2)	2 (1.1)	1 (0.6)
Nausea	35 (6.1)	2 (0.3)	16 (2.8)	0	6 (3.3)	0
Asthenia	33 (5.8)	7 (1.2)	19 (3.3)	2 (0.3)	2 (1.1)	0
Arthralgia	33 (5.8)	1 (0.2)	18 (3.1)	1 (0.2)	2 (1.1)	0

*TEAEs are reported as preferred terms (Medical Dictionary for Regulatory Activities v23.0) and were graded based on the National Cancer Institute Common Terminology Criteria for Adverse Events v4.03 by the investigator.

†TEAEs were reported for events that occurred during the period that patients were treated with placebo plus ADT and up to 30 days after the last dose, or up to the day before the start of open-label enzalutamide plus ADT, whichever was sooner.

ADT, androgen deprivation therapy; ENZA, enzalutamide; PBO, placebo; TEAE, treatment-emergent adverse event.

Table S4. Summary of TEAEs and Exposure-Adjusted TEAEs of Special Interest in the Placebo plus ADT Crossover Group (Safety Analysis Set)

	PBO + ADT Crossover			
	(n = 180)			
Median treatment duration, months (range)	23.9 (0.2–27.6)			
Total exposure, PY	312.3			
Any TEAE, no. (%)	149 (82.8)			
Any grade 3 or 4 TEAE, no. (%)	54 (30.0)			
Any TEAE leading to death, no. (%)	3 (1.7)			
Any study drug-related TEAE, no. (%)	91 (50.6)			
Any study drug-related TEAE leading to death, no. (%)	0			
Any TEAE of special interest, no. (%)	97 (53.9)			
	All Grades		Grade 3–4	
	Events		Events	
TEAE of special interest by group term*	no. (%)	(Rate)[†]	no. (%)	(Rate)[†]
Convulsions	1 (0.6)	1 (0.3)	0	0
Hypertension	14 (7.8)	14 (4.5)	6 (3.3)	6 (1.9)
Decreased neutrophil count	2 (1.1)	2 (0.6)	1 (0.6)	1 (0.3)

Cognitive/memory impairment	12 (6.7)	15 (4.8)	0	0
Ischemic heart disease	6 (3.3)	9 (2.9)	4 (2.2)	5 (1.6)
Other selected cardiovascular events	4 (2.2)	5 (1.6)	3 (1.7)	4 (1.3)
Posterior reversible encephalopathy syndrome	0	0	0	0
Fatigue	43 (23.9)	45 (14.4)	4 (2.2)	5 (1.6)
Renal disorders	0	0	0	0
Second primary malignancies	3 (1.7)	3 (1.0)	2 (1.1)	2 (0.6)
Falls	13 (7.2)	18 (5.8)	2 (1.1)	2 (0.6)
Fractures	12 (6.7)	17 (5.4)	2 (1.1)	3 (1.0)
Loss of consciousness	2 (1.1)	2 (0.6)	0	0
Thrombocytopenia	0	0	0	0
Musculoskeletal events	41 (22.8)	59 (18.9)	5 (2.8)	8 (2.6)
Severe cutaneous adverse reactions	0	0	0	0
Angioedema	0	0	0	0
Rash	6 (3.3)	7 (2.2)	1 (0.6)	1 (0.3)
Hepatic disorder	7 (3.9)	1 (4.2)	4 (2.2)	7 (2.2)

*TEAEs of special interest were based on prespecified combinations of preferred terms (Medical Dictionary for Regulatory Activities v23.0) and were graded based on the National Cancer

Institute Common Terminology Criteria for Adverse Events v4.03 by the investigator.

[†]Per 100 PYs of exposure.

ADT, androgen deprivation therapy; PBO, placebo; PY, patient-year; TEAE, treatment-emergent adverse event.

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3. Armstrong AJ, Shore ND, Szmulewitz RZ, et al: Efficacy of enzalutamide plus androgen deprivation therapy in metastatic hormone-sensitive prostate cancer by pattern of metastatic spread: ARCHES post hoc analyses. *J Urol* 205:1361-1371, 2021
4. Halabi S, Kelly WK, Ma H, et al: Meta-analysis evaluating the impact of site of metastasis on overall survival in men with castration-resistant prostate cancer. *J Clin Oncol* 34:1652-9, 2016